

Exercise 8 SQL Case Statements

Table: Products

Product-id	Product-name	Price
1	Laptop	1200.00
2	Phone	800.00
3	Keyboard	45.00
4	Monitor	300.00
5	Mouse	25.00

• Classify each product by price

• Expensive if price > 1000

• mid-range if price between 100 and 1000

• Budget if price < 100

SELECT

Product-name,

Price,

Case

WHEN price > 1000 THEN 'Expensive'

WHEN price BETWEEN 100 AND 1000 THEN 'Mid-Range'

ELSE 'Budget'

END AS price-category

FROM

products

Product-id	Product-name	Price
1	Laptop	1200
2	Phone	800.00
3	Keyboard	45.00
4	Monitor	300.00
5	Mouse	25.00

3. Table: orders

Order_id	Customer_name	amount
1	Alice	150
2	Bob	560
3	Charlie	999.99
4	Diana	45.50
5	Ethan	1200

Label each order:

- * High value for orders ≥ 1000
- * Medium value for $500 \leq 999.99$
- * Low value for orders < 500

Syntax: `SELECT customer_name, amount,`

`CASE`

`WHEN amount  $\geq 1000$  THEN 'High value'`

`WHEN amount  $\geq 500$  THEN 'medium value'`

`ELSE 'Low value'`

`END AS order_value_category`

`FROM orders`

Output	Customer_name	amount	order_value_category
	Alice	150	Low value
	Bob	560	medium value
	Charlie	999.99	medium value
	Diana	45.50	low value
	Ethan	1200	High value

5. Table employees

emp_id	emp_name	department	Salary
1	John	IT	85000
2	Sarah	HR	60000
3	Mark	IT	75000
4	Lucy	Finance	95000
5	Tom	HR	55000

• Categorize employee position

- If in IT and Salary > 80000 → Senior IT
- If in HR and Salary > 55000 → experienced HR
- otherwise → Staff

SELECT emp_name, department, Salary

CASE

WHEN department = 'IT' AND Salary > 80000 THEN 'Senior IT'

WHEN department = 'HR' AND Salary > 55000 THEN 'Experienced HR'

ELSE 'Staff'

END AS position_level

FROM employees;

Output'

Emp_Name	department	Salary	position_level
John	IT	85000	Senior IT
Sarah	HR	60000	experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

4 Table: Students

Student_id	Student_name	Score
1	Anna	92
2	Ben	76
3	Cara	59
4	David	83
5	Ella	68

• Assign a letter grade

• ≥ 90 : 'A'

80-89: 'B'

70-79: 'C'

60-69: 'D'

< 60: 'F'

SELECT Student_name, Score,

CASE

WHEN Score ≥ 90 THEN 'A'

WHEN Score ≥ 80 THEN 'B'

WHEN Score ≥ 70 THEN 'C'

WHEN Score ≥ 60 THEN 'D'

ELSE 'F'

END AS grade

FROM Students;

Output	Student_name	Score	grade
	Anna	92	A
	Ben	76	C
	Cara	59	F
	David	83	B
	Ella	68	D

5 Table: delivery

delivery_id

1

2

3

4

5

6

* label d

. ≤ 3

. 31-60

. 60

SELECT

Output

5 Table: deliveries

delivery-id	delivery-time-minutes
1	45
2	80
3	30
4	65
5	100
6	

• label delivery performance

• ≤ 30 mins: Fast

• 31-60 mins: On time

• 60 mins: late

SELECT delivery-id, delivery-time-minutes;

CASE

WHEN delivery-time-minutes ≤ 30 THEN 'Fast'

WHEN delivery-time-minutes ≤ 60 THEN 'on Time'

ELSE 'Late'

END AS performance

FROM deliveries

Output	delivery-id	delivery-time-minutes	performance
	1	45	on-time
	2	80	late
	3	30	Fast
	4	65	late
	5	100	late

6 Table tickets

ticket id	issue type	priority
1	login issue	1
2	Server down	3
3	Slow System	2
4	Email error	2
5	password reset	1

Convert priority to label

3 → High

2 → Medium

1 → Low

SELECT issue_type, priority,

CASE priority

WHEN 3 THEN 'High'

WHEN 2 THEN 'Medium'

WHEN 1 THEN 'Low'

END AS priority_label

FROM tickets

issue_type	Priority	priority_label
login issue	1	low
Server down	3	High
Slow System	2	medium
Email error	2	medium.
password reset	1	low

Output

1 Table emp

1 Table attendance

Student id	days present	total days
1	45	50
2	30	50
3	48	50
4	25	50
5	50	50

• Calculate attendance % and classify

• $\geq 90\%$ → Excellent

75-89% → Good

< 75% → Needs improvement

SELECT Student_id,

(days_present * 100.0 / total_days) AS attendance_percentage,

CASE

WHEN (days_present * 100.0 / total_days) $\geq$ 90 THEN 'Excellent'

WHEN (days_present * 100.0 / total_days) $\geq$ 75 THEN 'Good'

ELSE 'Needs Improvement'

END AS attendance_status

FROM attendance;

Output	Student_id	attendance_percentage	attendance_status
	1	90.0	excellent
	2	60.0	Needs Improvement
	3	96.0	excellent
	4	50.0	Needs Improvement
	5	100.0	excellent

8 Table : product - inventory

Product_id	Stock_qty
1	5
2	0
3	25
4	10
5	3

Label Stock status

0 → out of stock

1-5 → Low stock

5 → in stock

SELECT product_id, Stock_qty,

CASE

WHEN Stock_qty = 0 THEN 'out of stock'

WHEN Stock_qty BETWEEN 1 AND 5 THEN 'Low Stock'

ELSE 'in Stock'

END AS Stock_Status

FROM Product-Inventory;

Product_id	Stock_qty	Stock_Status
1	5	Low Stock
2	0	out of Stock
3	25	In Stock
4	10	In Stock
5	3	Low Stock

Out

9 Classify each class by the number of enrolled students

Table: Classes		
Class ID	Subject	Enrolled Students
1	Math	30
2	English	25
3	Science	15
4	Art	5
5	History	20

• Classify by size

$\geq 25 \rightarrow$ large

$10 - 24 \rightarrow$ medium

$< 10 \rightarrow$ small

Eligible To

SELECT Subject, enrolled_Students,

CASE

WHEN enrolled_Students ≥ 25 THEN 'Large'

WHEN enrolled_Students ≥ 10 THEN 'Medium'

ELSE 'Small'

END AS class_size_category

FROM classes;

ty

discount

Output	Subject	enrolled_Students	class_size_category
	Maths	30	large
	English	25	large
	Science	15	medium
	Art	5	small
	History	20	medium

discount

10 Table: payments

payment_id	amount	payment_method
1	50	Card
2	200	Cash
3	150	Card
4	75	Pay pal
5	300	Cash

• Apply discount flag

* If payment_method = Cash and amount $\geq 200 \rightarrow$ eligible

For discount

* otherwise $\rightarrow$ Not eligible

Query:
 SELECT Payment_id, payment_method, amount,
 CASE
 WHEN Payment_method = 'Cash' AND amount ≥ 200 THEN Eligible For
 Discount
 ELSE 'Not Eligible'
 END AS discount_eligibility
 FROM payments;

output	Payment_id	Payment_method	amount	discount_eligibility
	1	Card	50	Not eligible
	2	Cash	200	eligible For discount
	3	Card	150	Not eligible
	4	Pay pal	75	Not eligible
	5	Cash	300	eligible For discount

Out

9 Classifying each class enrolled student

Class id	Subject	enrolled_students
1	Math	30
2	English	25
3	Science	15
4	Art	5
5	History	20

• classifying by size

* $\geq 25 \rightarrow$ large

10 - 24 $\rightarrow$ medium

$< 10 \rightarrow$ small

SELECT Subject, enrolled_students
 CASE
 WHEN enrolled_students ≥ 25 THEN 'Large'
 WHEN enrolled_students < 25 AND ≥ 10 THEN 'Medium'
 ELSE 'Small'
 END AS size
 FROM classes;

Output	Subject	enrolled_students
	Maths	30
	English	25
	Science	15
	Art	5
	History	20